KANCHUKA INDURNGA

Software Engineer | Cybersecurity Researcher

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PROFESSIONAL SUMMARY

Software Engineering undergraduate (GPA 3.67/4.00) focused on cybersecurity and security automation. Hands-on experience building and deploying endpoint detection, cloud security, and vulnerability discovery tools using Python and modern security stacks. Strong in offensive and defensive security with a builder-first mindset.

EXPERIENCE

Lead Engineer | Early-Stage Service Platform | VÖID Junk Removal | 2025 Dec – Present

NYC-Based Service Tech Startup | Full-stack, AI, and Scalable Systems

- Designed and built a **programmatic SEO platform** generating **19,000+ scalable pages** across **1,000+ U.S. cities** and **1,700+ neighborhoods** using **Node.js**, with controlled service limits and intent-based CTAs to reduce thin-content and spam risk.
- Developed **client-side AI systems** using **TensorFlow.js (COCO-SSD)** for real-time junk detection (80+ object classes) and an **NLP-based chat-to-quote engine** converting natural language into instant pricing estimates, with **no server-side inference costs**.
- Engineered a **Progressive Web App (PWA)** with offline support, Service Workers, and installable manifests; launched **four free tools** (AI scanner, chat bot, donation calculator, moving countdown) to drive organic acquisition and differentiation.
- Built **hyper-local content pipelines** using structured city and neighborhood datasets (31,000+ entries), integrating landmarks, housing types, and competitor comparison pages to capture high-intent local and “vs” search queries.
- Implemented **technical SEO at scale**, including Schema.org JSON-LD (LocalBusiness, Service, FAQPage), per-state XML sitemaps, internal geo-linking, and performance optimization, achieving a **9.9/10 technical SEO audit score** and ~**90% CSS payload reduction**.

TECHNICAL SKILLS

Languages & Frameworks: Python (FastAPI, Pandas, Scikit-learn) Bash • JavaScript (Node.js, TensorFlow.js) • SQL
Offensive Security: Metasploit • Burp Suite • SQLMap • Custom Exploit Development • Web Application Pentesting • MITRE ATT&CK (T1055, T1059, T1110, T1486)
Defensive Security / SOC: Splunk • ELK Stack • Wazuh • Sigma Rules • YARA • Behavioral Detection • Threat Hunting
Malware Analysis & Forensics: Cuckoo Sandbox • IDA Pro • Ghidra • Wireshark • Suricata • Sysmon • Volatility • Autopsy
DevSecOps & Automation: Docker • CI/CD (GitHub Actions, GitLab CI) SAST/DAST Integration • Infrastructure as Code

SOFT SKILLS

Strategic Thinking: System design planning, long-term project planning, basic threat modeling, task prioritization, and risk awareness in technical decisions.
Problem Solving: Identifying root causes, structured troubleshooting, debugging issues step by step, and making decisions based on evidence.
Communication: Writing clear technical documentation, explaining security and technical issues to non-experts, and collaborating with technical and non-technical teams.
Adaptability: Learning new tools quickly, working across different technical areas, and adjusting workflows based on project and team needs.

FEATURED TECHNICAL PROJECTS

ZoEDR Endpoint Detection & Response *github.com/RicheByte/ZoEDR*

- Deployed ZoEDR on 3+ Linux endpoints for continuous threat monitoring, collected real-time telemetry, and generated structured alerts using behavioral heuristics; implemented automated quarantine and recovery actions reducing manual response time by 45%

fsocietyOS: Modular Penetration Testing Framework | *Python, Bash,* *github.com/RicheByte/fsocietyOS*

- Engineered a unified CLI framework consolidating 50+ reconnaissance and attack scripts across 6 domains, eliminating tool-switching friction.
- Automated vulnerability discovery for SQLi, XSS, and LFI, cutting exploit-to-discovery time by **90%** during simulated engagements.
- Designed a modular plugin architecture allowing for rapid integration of new exploit modules without core code alteration.

CyberSamantha: AI-Powered Security Knowledge Assistant | *Python, LLMs* *github.com/RicheByte/cyberSamantha*

- Built a Retrieval-Augmented Generation (RAG) system querying a 10,000+ chunk vector store, delivering context-aware security answers in under **2 seconds**.

- Optimized the knowledge ingestion pipeline, reducing data parsing time from **40 hours to 10 minutes** for unstructured security documentation.

CloudMonkey: CSPM & Audit Tool | *Python, AWS API, SQLite* *github.com/RicheByte/cloudMonkey*

- Developed a Cloud Security Posture Management (CSPM) CLI to instantly detect critical misconfigurations (Public S3, exposed. env).
- Reduced Mean Time to Remediation (MTTR) by **60%** for cloud asset audits by providing actionable, prioritized findings.
- Built an audit logging system using SQLite to maintain an immutable record of findings across 5,000+ scanned assets.

BugHunter: Threat Intelligence Aggregator | *Python, Scikit-learn, JSON* *github.com/RicheByte/bugHunter*

- Consolidated threat data from NVD, Exploit-DB, and GitHub into a unified JSONL feed, reducing audit reporting time by **80%**.
- Integrated a Machine Learning model (Scikit-learn) to predict vulnerability severity, achieving **92% accuracy** in identifying potential zero-day patterns.

Security Auditor Plugin: SAST/SCA Engine | *Python, Git Analysis* *github.com/RicheByte/beforeAuditor*

- Created a 7-in-1 static analysis engine combining secret detection, dependency checking, and SAST into a single **2.34-second** scan.
- Engineered an offline zero-day detection algorithm using differential git analysis and fuzzy hashing to identify high-confidence vulnerabilities.

RESEARCH & PUBLICATIONS

Pentester Playbook v3.0 | *Comprehensive Red Team Framework* *richebyte.github.io/pentesterplaybook*

- Standardized 200+ attack techniques across critical vectors including Active Directory (Kerberos/Golden Tickets), Cloud Infrastructure, and ICS/SCADA protocols.

Malware vs. Malware Ecology: Interactions Within the Wild | *Research Study* *github.com/RicheByte/vigilanteMalware*

- Authored a study exploring "digital biology," analyzing how active malware strains compete for system resources and interfere with rival processes.

Malware101 | Educational Content on Analysis *github.com/RicheByte/Malware101*

- Developed a comprehensive library of **100+ hand-coded malware samples** (including Process Injection, Rootkits, and Fileless Malware) within a secured, GPG-encrypted research environment to demonstrate advanced evasion, persistence, and defensive analysis techniques. All research conducted in isolated environments for defensive education.

EDUCATION

BSc (Hons) in Software Engineering | Seagis Campus
GPA: 3.67 / 4.00 (*Expected Graduation: 2027)*
Key Coursework: Network Security, Data Structures & Algorithms, Ethical Hacking, Database Systems.
Bachelor of Science in Business Administration (BSBA) | University of the People
Expected Graduation: 2028
Key Coursework: Entrepreneurship, Business and Society, Quality Management, Principles of Business Management

CERTIFICATIONS & EXTRACURRICULARS

CompTIA Security+ (Expected 08/2026)
Published Author: Technical breakdown of articles featured on *hacker.super.site*
Leadership: Led multidisciplinary engineering teams in agile environments in the university; authored documentation for open-source frameworks.
Planning & Development Director | ICT Club, Saegis Campus(Jan 2026 - present) : Led the planning and technical execution of campus-wide hackathons, ideathons, and CTF competitions while developing the ICT Club's official website for event publishing, member onboarding, and internal coordination, and serving as the technical bridge between students, faculty, and external mentors to deliver structured, industry-relevant events.